

Assessment Annexure C - Questions & Answers

Question 1 (10 Marks)

Question: Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study?

Age of child	A	B	C
5-6 years	18	22	20
7-8 years	2	28	40
9-10 years	20	10	40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

Answer 1:

Part 1: Chi-Square Test of Independence

- **Null Hypothesis (\$H_0\$):** Photograph preference is independent of age group.
- **Alternative Hypothesis (\$H_1\$):** Photograph preference depends on age group.
- **Degrees of Freedom:** $df = (3 - 1) \times (3 - 1) = 4$
- **Calculated Chi-Square (\$\chi^2\$):** 28.303
- **Critical Value (\$\chi^2_{\alpha=0.05, df=4}\$):** 9.488
- **p-value:** 0.0000108 (1.08×10^{-5}) ($0.0000108 < 0.05$)
- **Conclusion:** Reject \$H_0\$. The data suggests a statistically significant relationship between age and photograph preference.

Part 2: What is wrong with this study?

1. **Confounding Visual Variables:** Photographs A, B, and C differ in multiple unmeasured adult attributes (gender, facial expression, background, perceived attractiveness/trustworthiness), making it impossible to isolate the cause of preference.
2. **Forced-Choice Categorical Format:** Forcing children to pick only one photo generates aggregated counts rather than continuous rating scores, losing preference intensity.
3. **Potential Social Desirability / Stereotyping:** Children's choices may be influenced by familiarity or perceived authority figures rather than age-specific cognitive differences.
4. **Sample Imbalance:** Group 5-6 years ($n=60$) has fewer participants than 7-8 and 9-10 years ($n=70$ each).

Part 3: R Function Calculations (cov & cor)

```
# Data definition
prefs <- data.frame(A = c(18, 2, 20), B = c(22, 28, 10), C = c(20, 40, 40))

# Sample covariance between B and C:
cov(prefs$B, prefs$C)
# Output: -30

# Sample covariance matrix:
cov(prefs)
# Output:
#      A      B      C
# A 100.00 -30.00 -70.00
```

```
# B -30.00 84.00 -30.00
# C -70.00 -30.00 133.33

# Sample correlation between B and C:
cor(prefs$B, prefs$C)
# Output: -0.2835

# Sample correlation matrix:
cor(prefs)
# Output:
#      A      B      C
# A  1.0000 -0.3273 -0.6061
# B -0.3273  1.0000 -0.2835
# C -0.6061 -0.2835  1.0000
```

Question 2 (10 Marks)

Question: Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot?

Player ID	Player Name	Points Scored	Player ID	Player Name	Points Scored
P1	Player A	12	P9	Player I	11
P2	Player B	15	P10	Player J	35
P3	Player C	14	P11	Player K	16
P4	Player D	10	P12	Player L	17
P5	Player E	18	P13	Player M	19
P6	Player F	20	P14	Player N	21
P7	Player G	22	P15	Player O	23
P8	Player H	13			

Answer 2:

- Order Data:** \$10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35\$ (\$N = 15\$)
- Calculate Quartiles:**
 - **Median (\$Q_2\$):** \$17\$ (8th value)
 - **First Quartile (\$Q_1\$):** \$13\$ (Median of lower 7 values)
 - **Third Quartile (\$Q_3\$):** \$21\$ (Median of upper 7 values)
 - **Interquartile Range (\$IQR\$):** \$Q_3 - Q_1 = 21 - 13 = 8\$
- Calculate Outlier Fences:**
 - **Lower Fence:** \$Q_1 - 1.5 \text{ times } IQR = 13 - 1.5(8) = 1\$
 - **Upper Fence:** \$Q_3 + 1.5 \text{ times } IQR = 21 + 1.5(8) = 33\$
- Outlier Decision:** Player J (35 points) is greater than the Upper Fence (\$35 > 33\$). Therefore, **Player J (35 pts) is an outlier.**

Boxplot Tennis Players

Question 3 (10 Marks)

Question: A bank wants to decide whether to approve a loan based on customer attributes (Age, Income, Employment Status, Credit Score). Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No).

Answer 3:

1. System Entropy: Total $S = 15$ instances (9 ext{ Yes}, 6 ext{ No}).

$$S = - \left(\frac{9}{15} \log_2 \frac{9}{15} \right) - \left(\frac{6}{15} \log_2 \frac{6}{15} \right) = 0.9710 \text{ bits}$$

2. Information Gain for Attributes:

- **Employment Status:** Weighted Entropy = 0.4764 $\rightarrow$ **Gain = 0.4946 (Highest $\rightarrow$ Root Node)**
- **Credit Score:** Gain = 0.4157
- **Income:** Gain = 0.3700
- **Age:** Gain = 0.0268

3. Decision Tree Rules:

```
[Root Node: Employment Status]
├─ unemployed --> DECISION = NO (5 No, 0 Yes)
├─ self-employed
│   ├─ If Income == low OR Credit Score == poor --> DECISION = NO
│   └─ Else --> DECISION = YES
└─ employed
    ├─ If Credit Score == poor --> DECISION = NO
    └─ Else --> DECISION = YES
```

Question 4 (5 Marks)

Question: Two Maths teachers are comparing Year 9 exam results:

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

- (i) Find which class scored higher mean, median and range.
(ii) Plot in boxplot and give inferences.

Answer 4:

(i) Statistical Measures Table:

Metric	Class A Calculation	Class A Result	Class B Calculation	Class B Result	Higher Class
Mean	$592 / 9$	65.78	$559 / 9$	62.11	Class A
Median	Middle (5th)	66.00	Middle (5th)	60.00	Class A
Range	$95 - 35$	60.00	$84 - 50$	34.00	Class A

Result: Class A scored higher in Mean, Median, and Range.

(ii) Boxplot & Inferences:

Class Comparison Boxplot

- **Central Tendency:** Class A performed better overall with higher median (66) and mean (65.78).

- **Variability:** Class B showed significantly higher score consistency (Range = \$34\$), whereas Class A scores had high dispersion (Range = \$60\$).

Question 5 (10 Marks)

Question: Hospital age and body fat data for 18 adults:

- Calculate mean, median, standard deviation of age and %fat.
- Draw boxplots for age and %fat.
- Draw a scatter plot and a q-q plot based on these two variables.

Answer 5:

(a) Summary Statistics:

Variable	Mean	Median	Standard Deviation
Age (years)	46.44	51.00	13.22
Body Fat (%fat)	28.78	31.30	9.25

(b) Boxplots for Age and %Fat:

Q5 Boxplots

(c) Scatter Plot and Q-Q Plot:

Q5 Scatter and QQ plot

- **Scatter Plot:** Strong positive linear relationship ($r \approx 0.82$). As age increases, %fat increases.
- **Q-Q Plot:** Points align closely with the normal reference line, indicating %fat is reasonably normally distributed.